



▲ TIDE POOLS

On rocky shores, the receding tide often leaves pools of water among the rocks. When this happens high on the shore, the pools heat up and may dry out, but pools that form below midtide level are not exposed to the Sun for so long, so they stay cool. They make ideal refuges for marine life that cannot survive being left high and dry for several hours, and the biggest tide pools may even contain large fish.



▲ BEACH LIFE

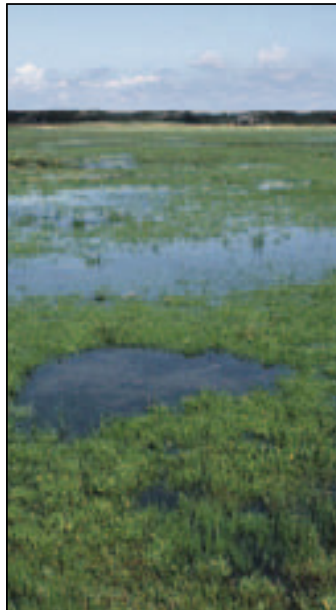
Most of the animals that live on sandy beaches and tidal mudflats are burrowing clams and worms, which use tubes and tentacles to gather food from the water at high tide. There may be millions of them, but when the tide goes out they retreat into their burrows and are hidden from view. The shorebirds that feed at low tide know their hiding places, however, and are experts at digging them out.

COASTAL PLANTS



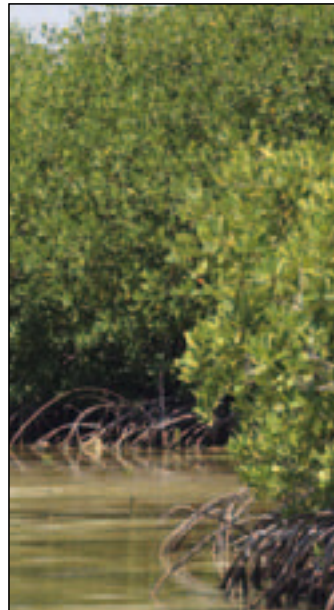
ROCKY CLIFFS

In the splash zone above the level of high spring tides, the shore is colonized by flowering plants such as thrift. These maritime plants can tolerate being drenched in salt spray, which would kill most species. This enables them to spread over large areas of the shore where other plants cannot grow. Although the soil is usually very shallow and poor, they can often flourish in the few pockets of fertile earth that build up in crevices and on rocky cliff ledges.



SALT MARSHES

In cooler regions, the upper shores of sheltered river estuaries are overrun by low-growing plants such as sea lavender and cord grass. These plants are unusual because they can survive being flooded by salt water at high tide. Each plant species has a different tolerance to this, so they grow in well marked zones above midtide level. They form broad, muddy salt marshes, which are important refuges and feeding sites for shorebirds and other animals.



MANGROVES

The tropical equivalent of a salt marsh is a mangrove swamp—a forest of salt-tolerant trees that are specially adapted to grow in tidal mud. They have modified roots that rise above the mud to absorb oxygen from the air. The roots also trap mud suspended in the water, enabling mangroves to spread along coasts. They provide habitats for a variety of animals from baby fish to crocodiles, besides forming natural barriers to storm waves.



SEAGRASS MEADOWS

Seagrasses are the only flowering plants that live underwater. They grow on sandy seabeds in sheltered, shallow water, in tidal estuaries, and coral reef lagoons. They are an important source of food for sea turtles, the strange "sea cows" known as dugongs, and—at low tide—grazing birds such as geese. Seagrass meadows also provide vital refuge for small fish, such as this seahorse, as well as the young of bigger fish that live in deeper water as adults.